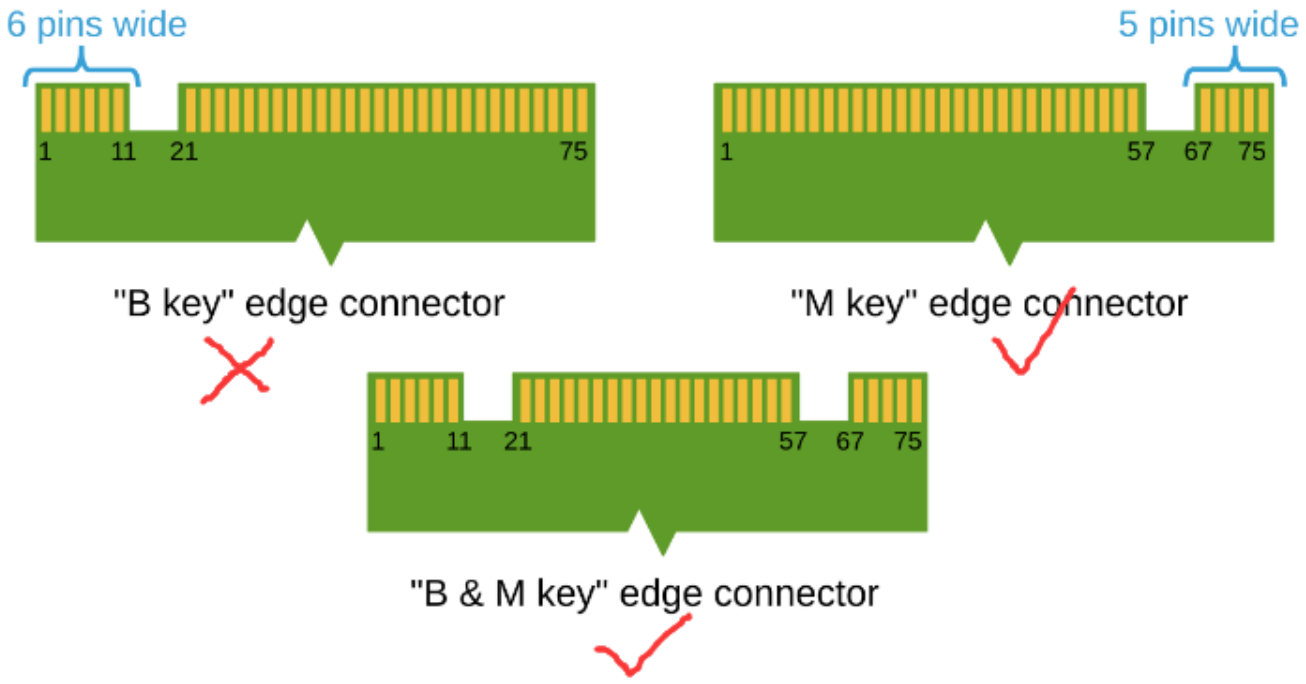
10. Flash the system onto your own SSD

10. Flash the system onto your own SSD

1. Preparation before flashing
2. Flash the Yahboom factory system
    1. Flash the SUPER bootloader
    2. Flash the Yahboom factory image onto the SSD
        2.1 Format the SSD
            1. Download DiskGenius
        2.1.2. Use DiskGenius
            2.1.2.1, Delete partition
            2.1.2.2. Create a new partition
    2.2 Restore the factory image
        2.2.1. Install Win32DiskImager
        2.2.2 Use Win32DiskImager

1. Preparation before flashing

A self-supplied SSD only supports M.2 Key M and M.2 Key B+M interface storage devices, and is not compatible with M.2 Key B interface storage devices. The storage capacity must not be lower than 256G.



The recommended size is 2280, 22X80(mm)



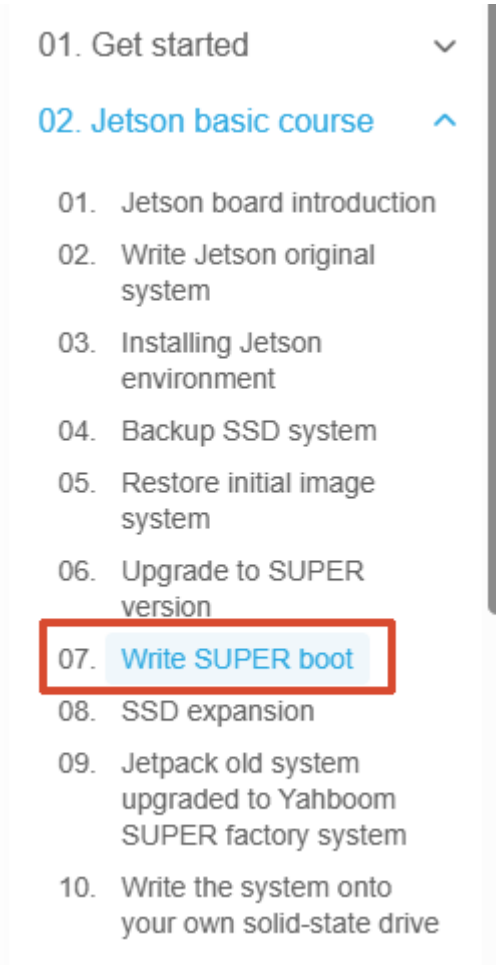
2. Flash the Yahboom factory system

Notes on using the Yahboom factory image: to boot the mainboard, two conditions must be met — one is flashing the super bootloader, and the other is flashing the Yahboom factory image onto the SSD. Neither can be missing, otherwise the board will not boot. (Must read)

1. Flash the SUPER bootloader

Note: If you purchased the SUB kit, the super bootloader has already been flashed at the factory, so you do not have to flash the super bootloader and you can skip this step.

To flash the bootloader, you can refer directly to the [**07.Write SUPER boot (Must read)**] tutorial.



2. Flash the Yahboom factory image onto the SSD

Before flashing, put the SSD into the SSD enclosure.





2.1 Format the SSD

1. Download DiskGenius

Download URL: https://www.diskgenius.com/

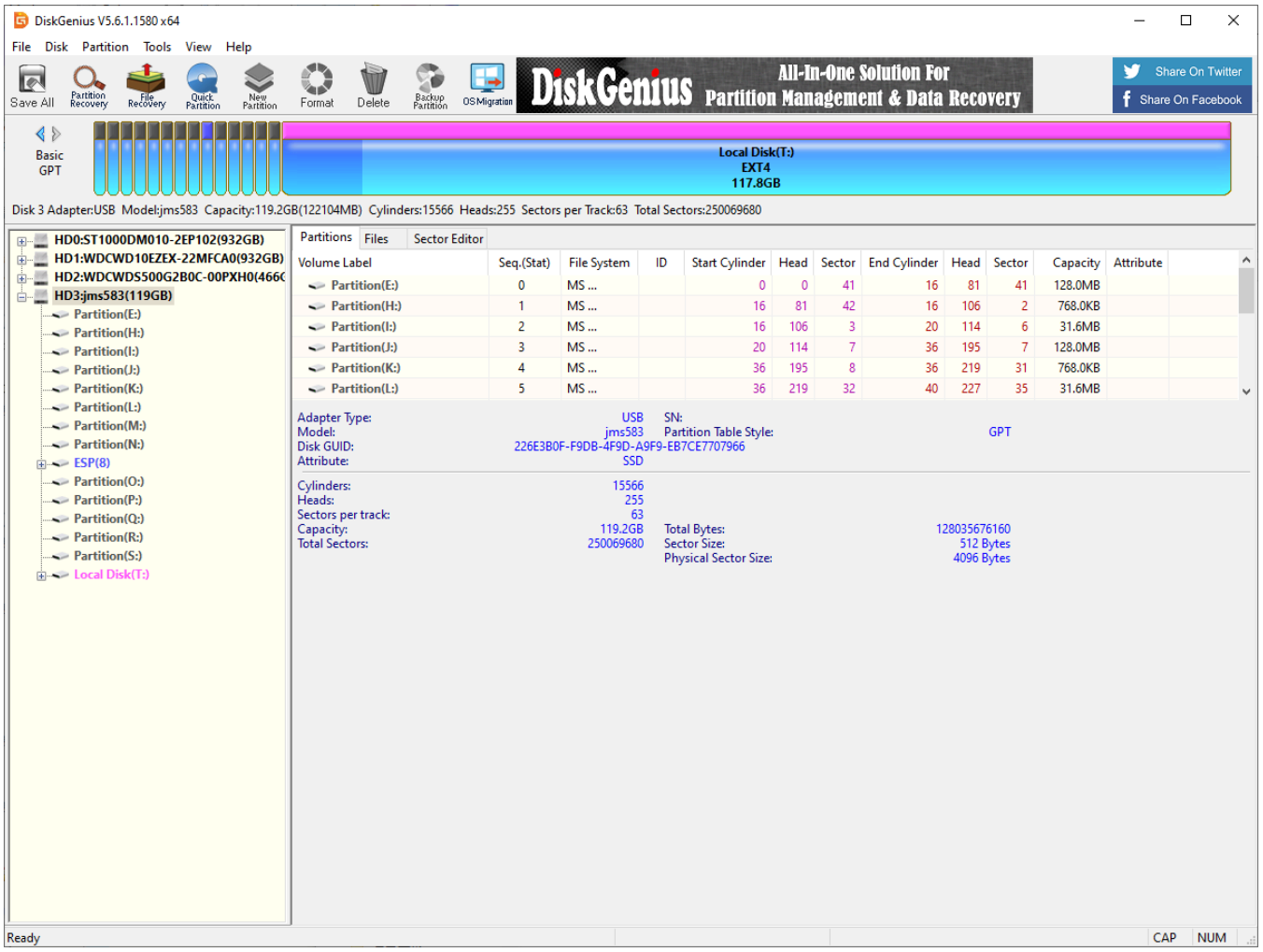


What Can DiskGenius Do?

Data Recovery
Recover deleted, lost, or formatted data from hard drives, USB drives, SD cards, RAW drives, RAID, and more, with advanced recovery options.

Partition Manager
Resize, format, create, hide, and wipe partitions. Convert disks between GPT and MBR, repartition HDDs/SSDs, allocate free space, and other features.

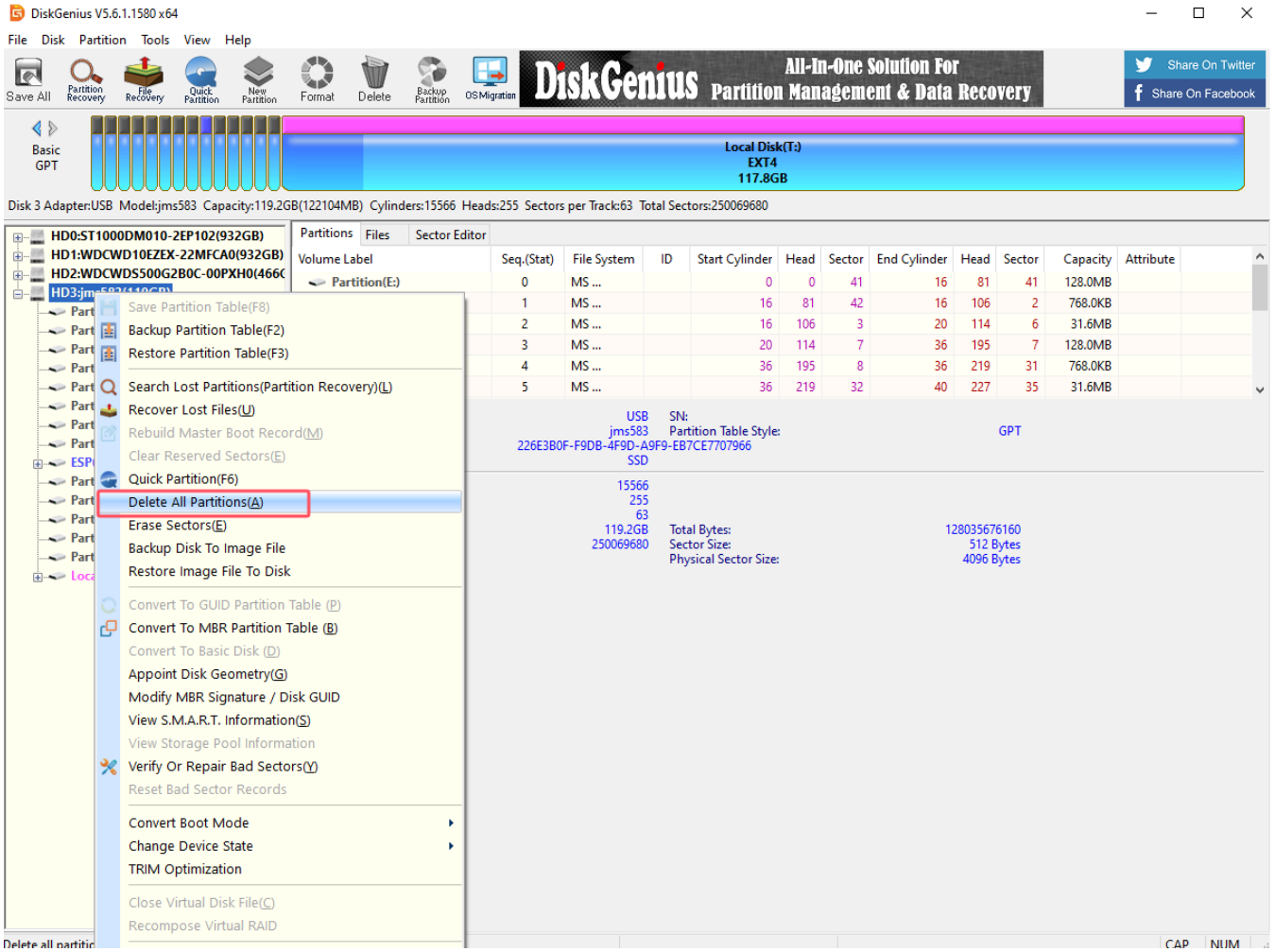
Extract the downloaded archive and you can open `DiskGenius.exe` to use it:

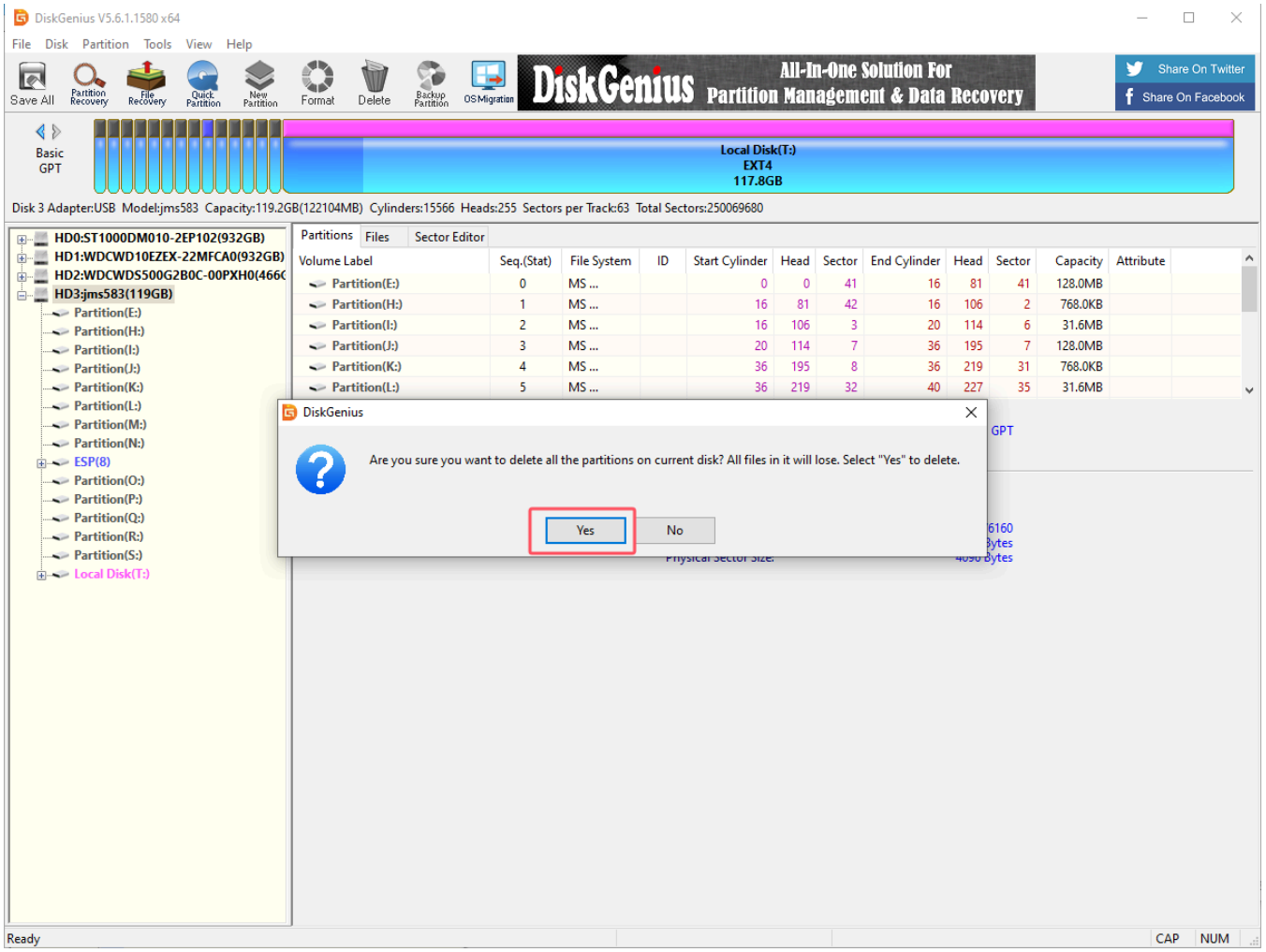


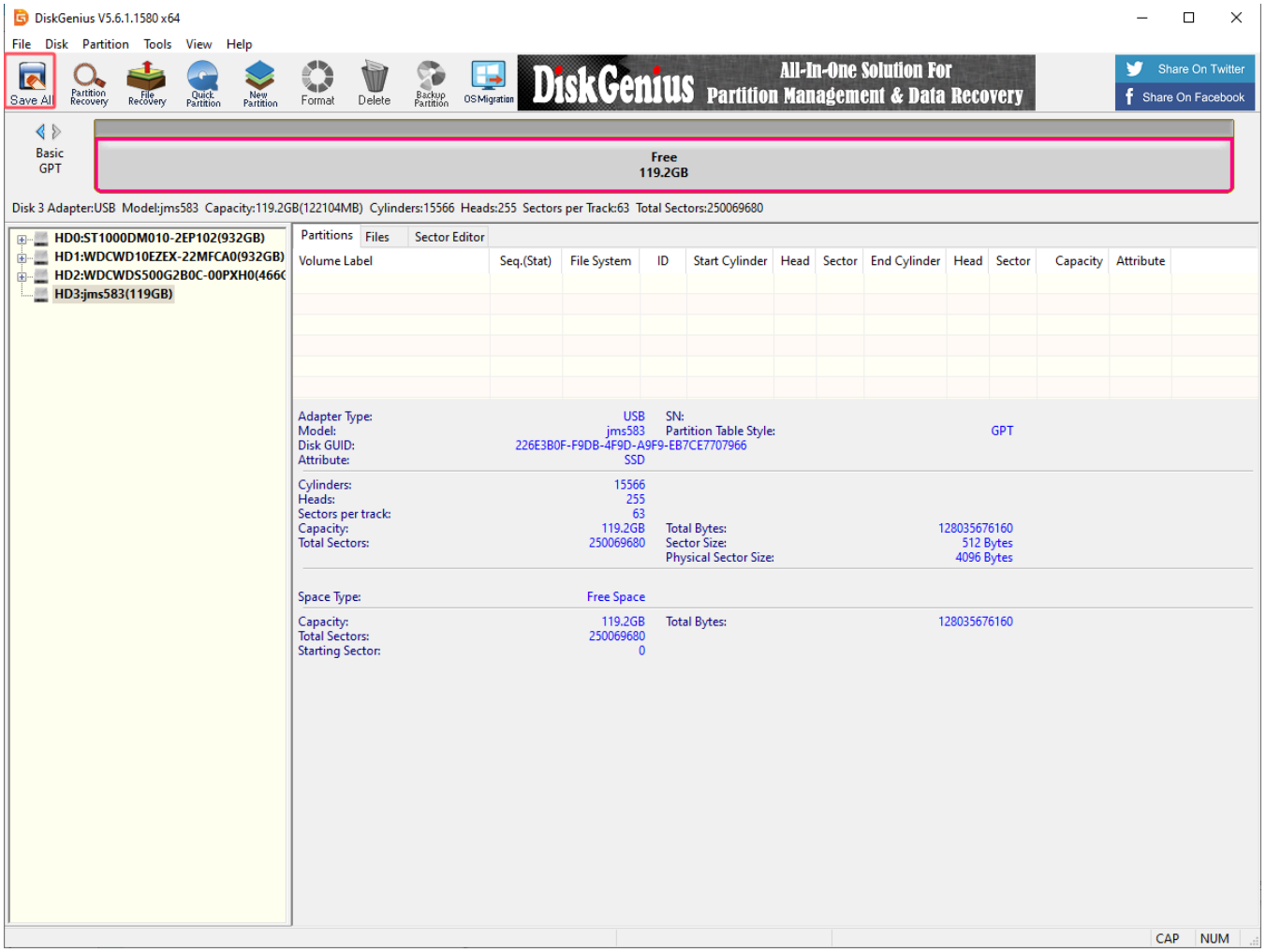
2.1.2. Use DiskGenius

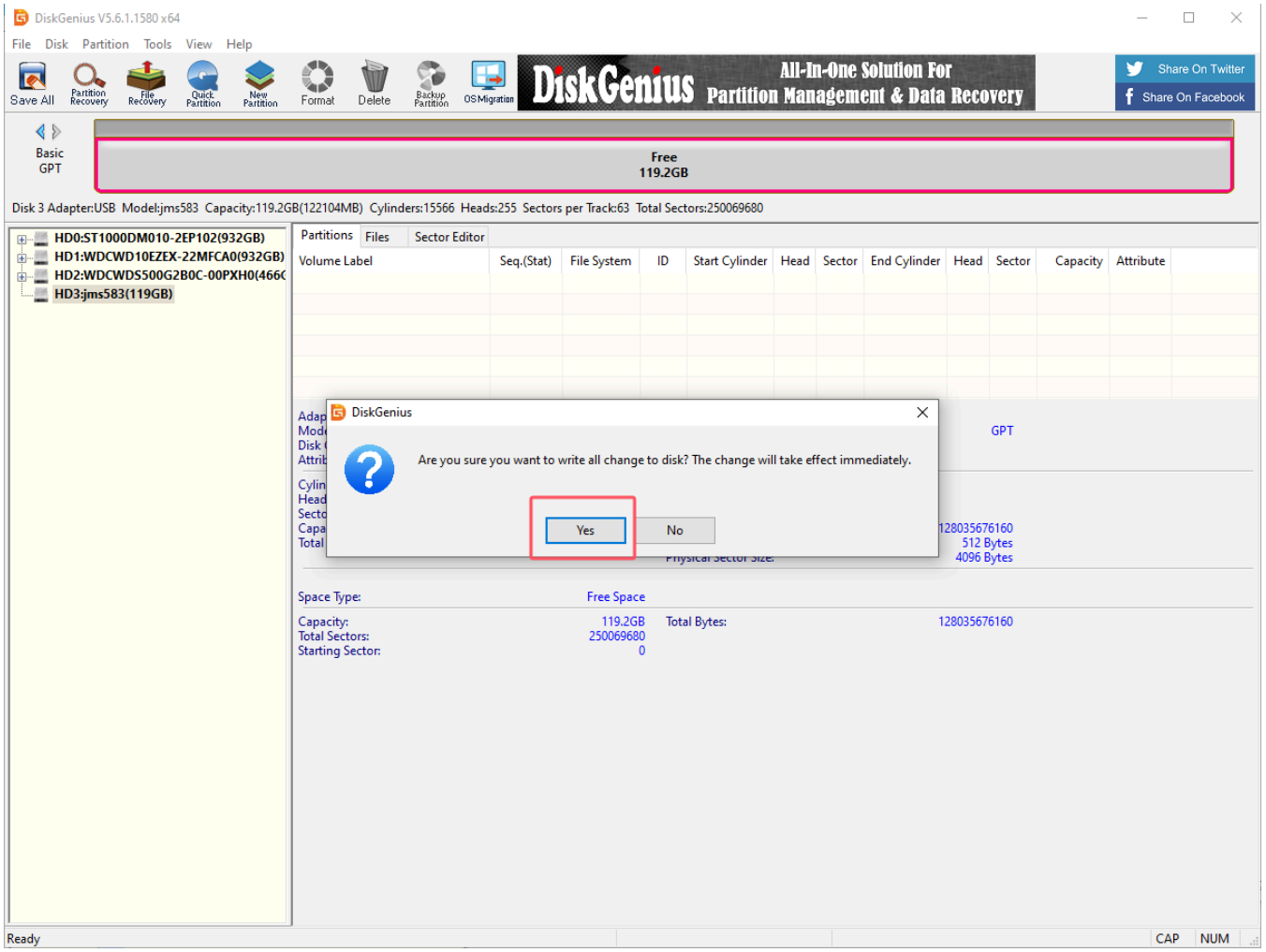
2.1.2.1, Delete partition

Deleting a partition will clear the disk data. Please confirm whether the drive letter is the disk that needs to be formatted before confirming the operation: you can judge based on the disk size and the newly added drive letter of the connected disk




DiskGenius V5.6.1.1580 x64
File Disk Partition Tools View Help
Save All
Partition Recovery
File Recovery
Quick Partition
New Partition
Format
Delete
Backup Partition
OS Migration
DiskGenius
All-In-One Solution For
Partition Management & Data Recovery
Share On Twitter
Share On Facebook
Basic
GPT
Local Disk(T:)
EXT4
117.8GB
Disk 3 Adapter:USB Model:jms583 Capacity:119.2GB(122104MB) Cylinders:15566 Heads:255 Sectors per Track:63 Total Sectors:250069680
HD0:ST1000DM010-2EP102(932GB)
HD1:WDCWD10EZEX-22MFCA0(932GB)
HD2:WDCWDS500G2B0C-00PXH0(466(
HD3:jms583(119GB)
Partition(E:)
Partition(H:)
Partition(I:)
Partition(J:)
Partition(K:)
Partition(L:)
Partition(M:)
Partition(N:)
ESP(8)
Partition(O:)
Partition(P:)
Partition(Q:)
Partition(R:)
Partition(S:)
Local Disk(T:)
Partitions
Files
Sector Editor
Volume Label
Seq.(Stat)
File System
ID
Start Cylinder
Head
Sector
End Cylinder
Head
Sector
Capacity
Attribute
Partition(E:) 0 MS ... 0 0 41 16 81 41 128.0MB
Partition(H:) 1 MS ... 16 81 42 16 106 2 768.0KB
Partition(I:) 2 MS ... 16 106 3 20 114 6 31.6MB
Partition(J:) 3 MS ... 20 114 7 36 195 7 128.0MB
Partition(K:) 4 MS ... 36 195 8 36 219 31 768.0KB
Partition(L:) 5 MS ... 36 219 32 40 227 35 31.6MB
DiskGenius
Are you sure you want to delete all the partitions on current disk? All files in it will lose. Select "Yes" to delete.
Yes
No
GPT
Ready
CAP
NUM

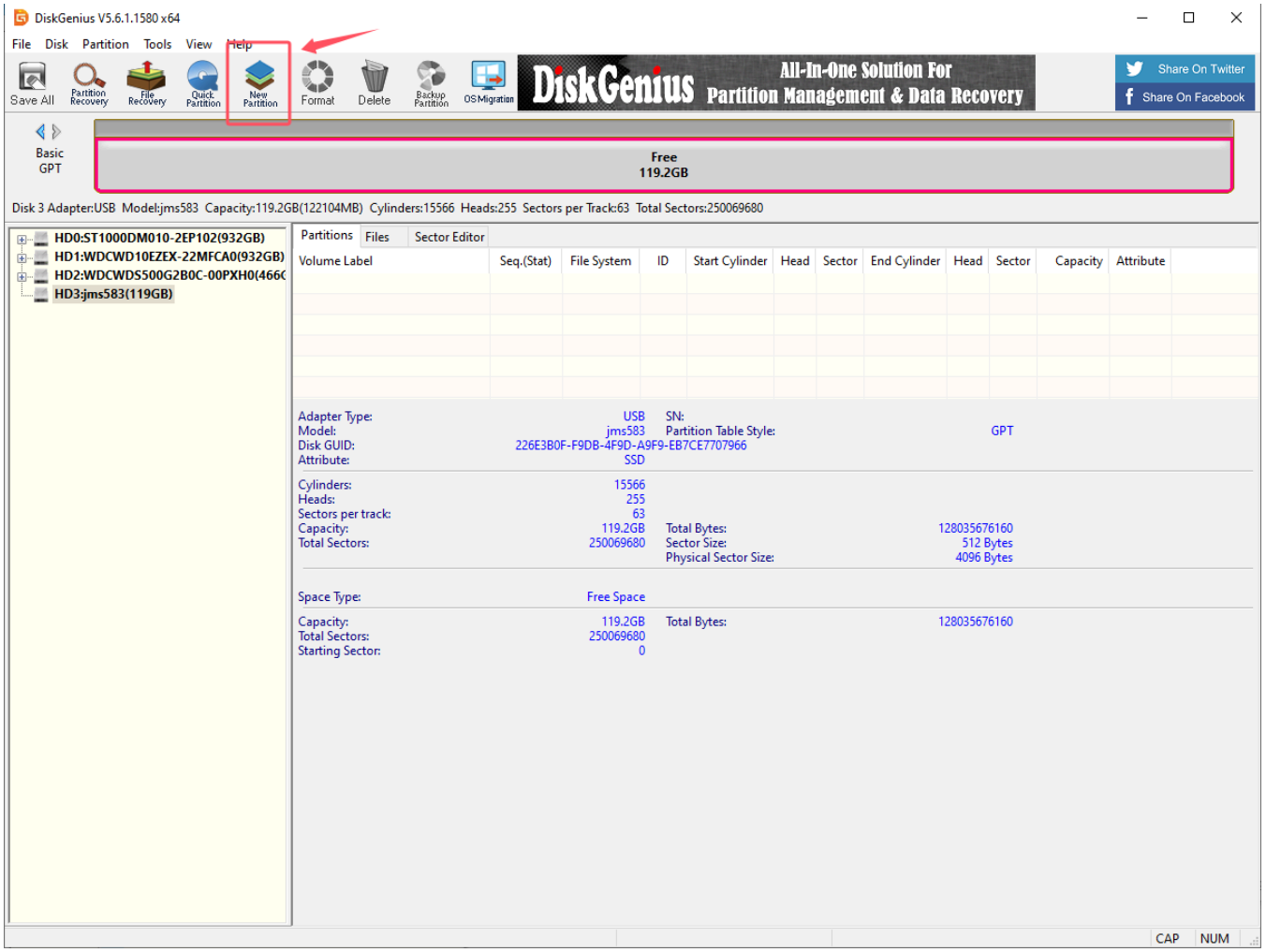

DiskGenius V5.6.1.1580 x64
File Disk Partition Tools View Help
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File Recovery
Quick Partition
New Partition
Format
Delete
Backup Partition
OS Migration
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Share On Twitter
Share On Facebook
Basic GPT
Free
119.2GB
Disk 3 Adapter:USB Model:jms583 Capacity:119.2GB(122104MB) Cylinders:15566 Heads:255 Sectors per Track:63 Total Sectors:250069680
HD0:ST1000DM010-2EP102(932GB)
HD1:WDCWD10EZEX-22MFCA0(932GB)
HD2:WDCWDS500G2B0C-00PXH0(466(
HD3:jms583(119GB)
Partitions
Files
Sector Editor
Volume Label
Seq.(Stat)
File System
ID
Start Cylinder
Head
Sector
End Cylinder
Head
Sector
Capacity
Attribute
Adapter Type: USB
Model: jms583
Disk GUID: 226E3B0F-F9DB-4F9D-A9F9-EB7CE7707966
Attribute: SSD
SN:
Partition Table Style: GPT
Cylinders: 15566
Heads: 255
Sectors per track: 63
Capacity: 119.2GB
Total Sectors: 250069680
Total Bytes: 128035676160
Sector Size: 512 Bytes
Physical Sector Size: 4096 Bytes
Space Type: Free Space
Capacity: 119.2GB
Total Sectors: 250069680
Starting Sector: 0
Total Bytes: 128035676160
CAP
NUM

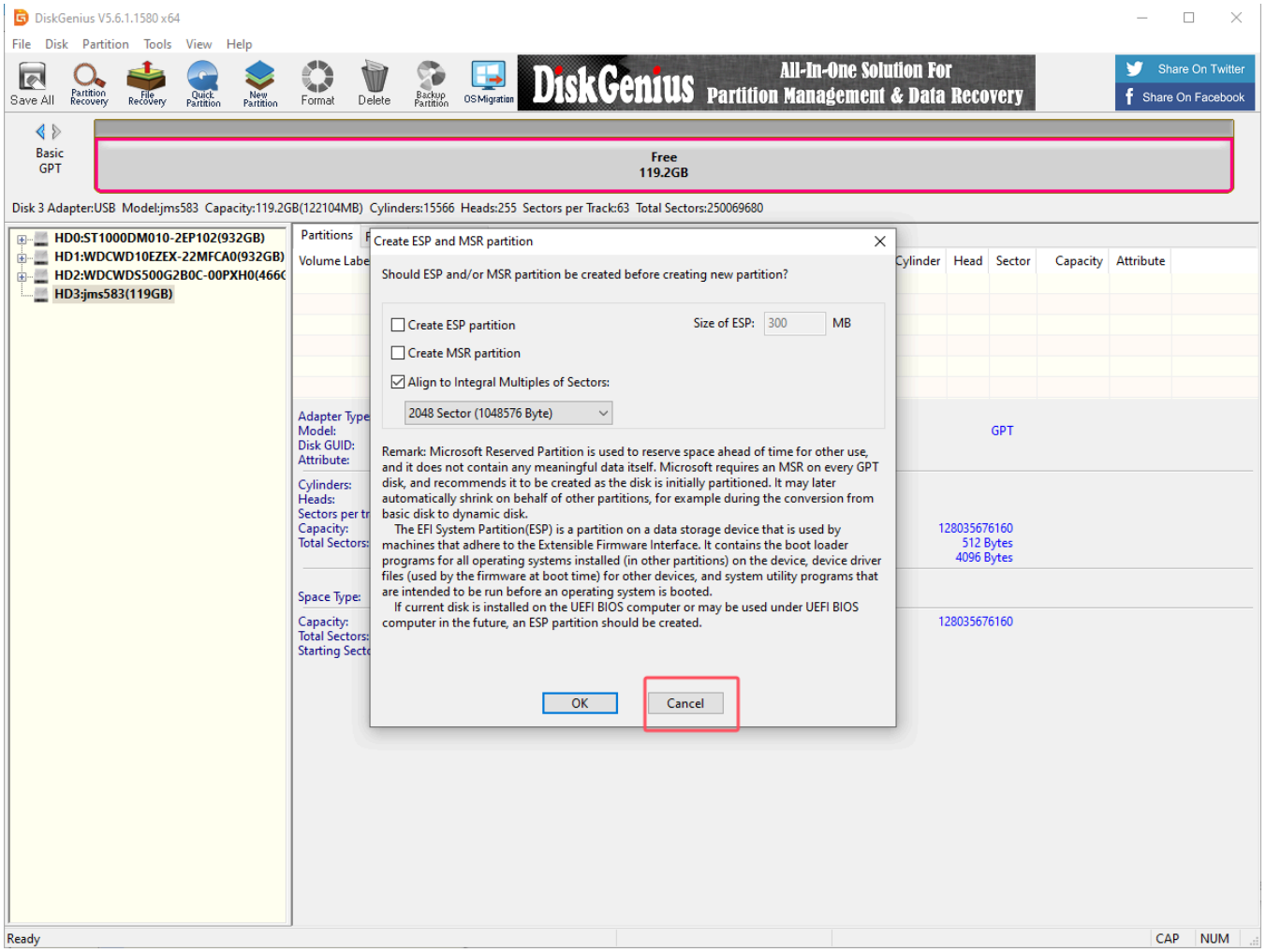


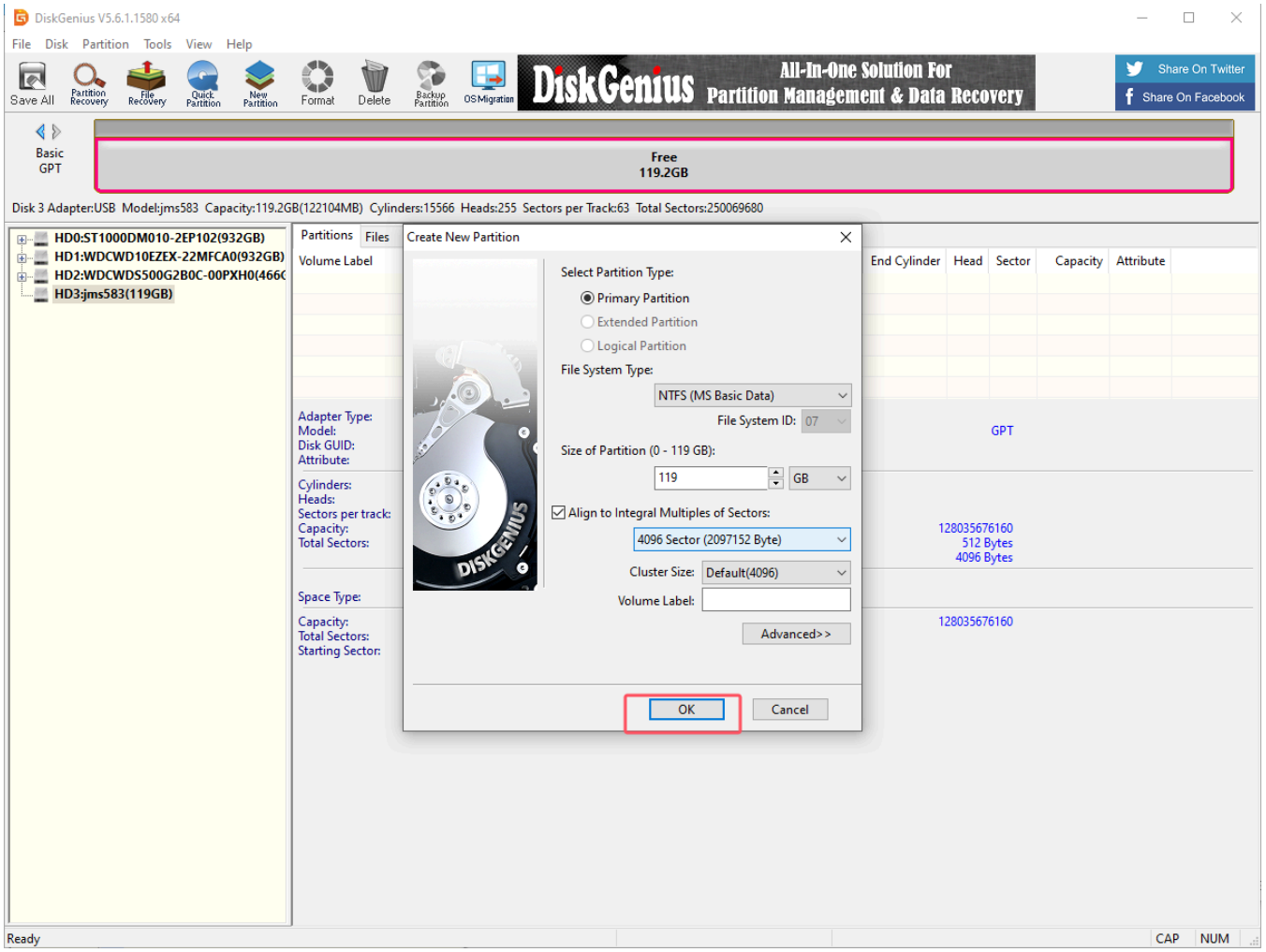
2.1.2.2. Create a new partition

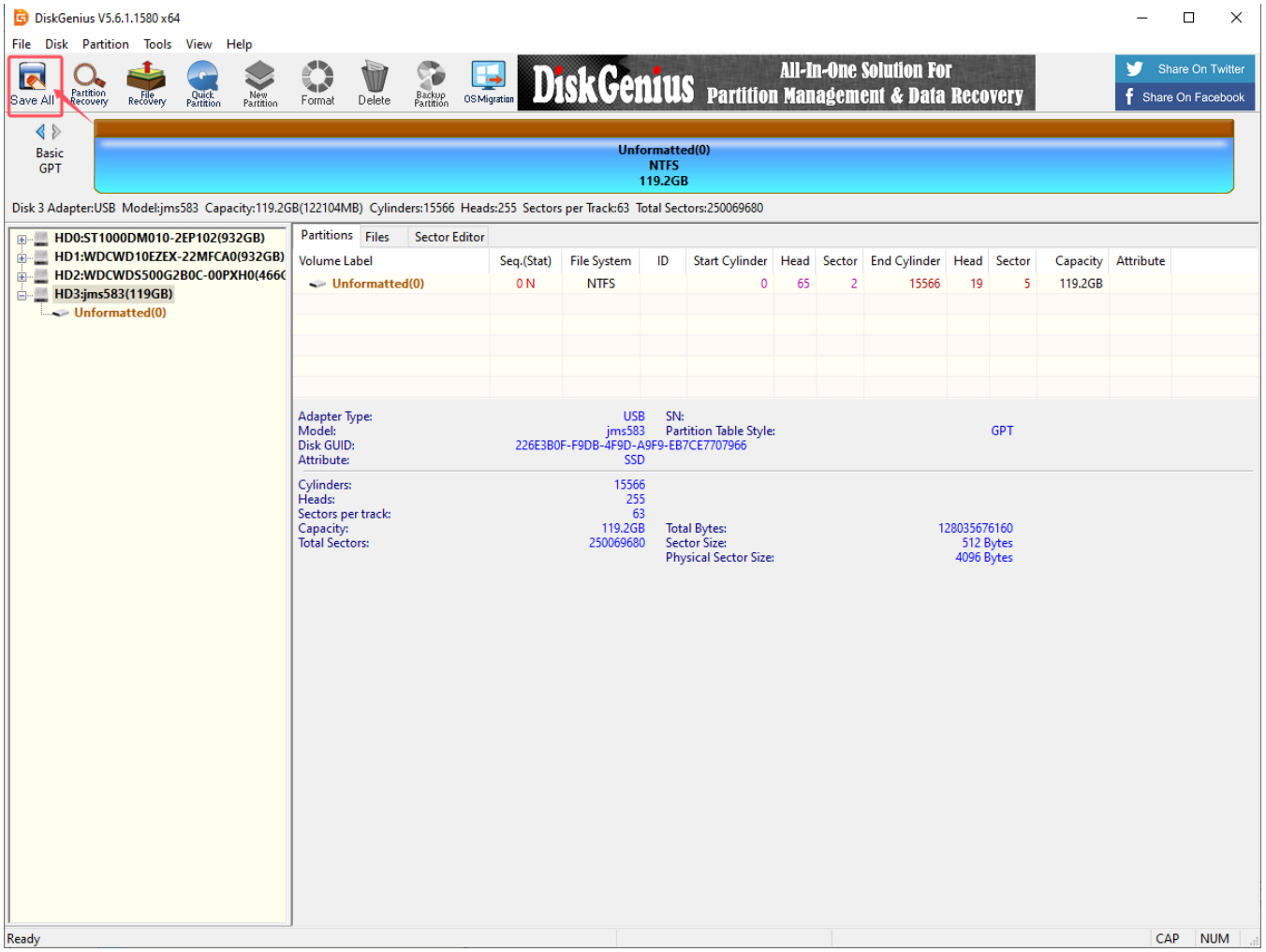
Partition the SSD into NTFS format.

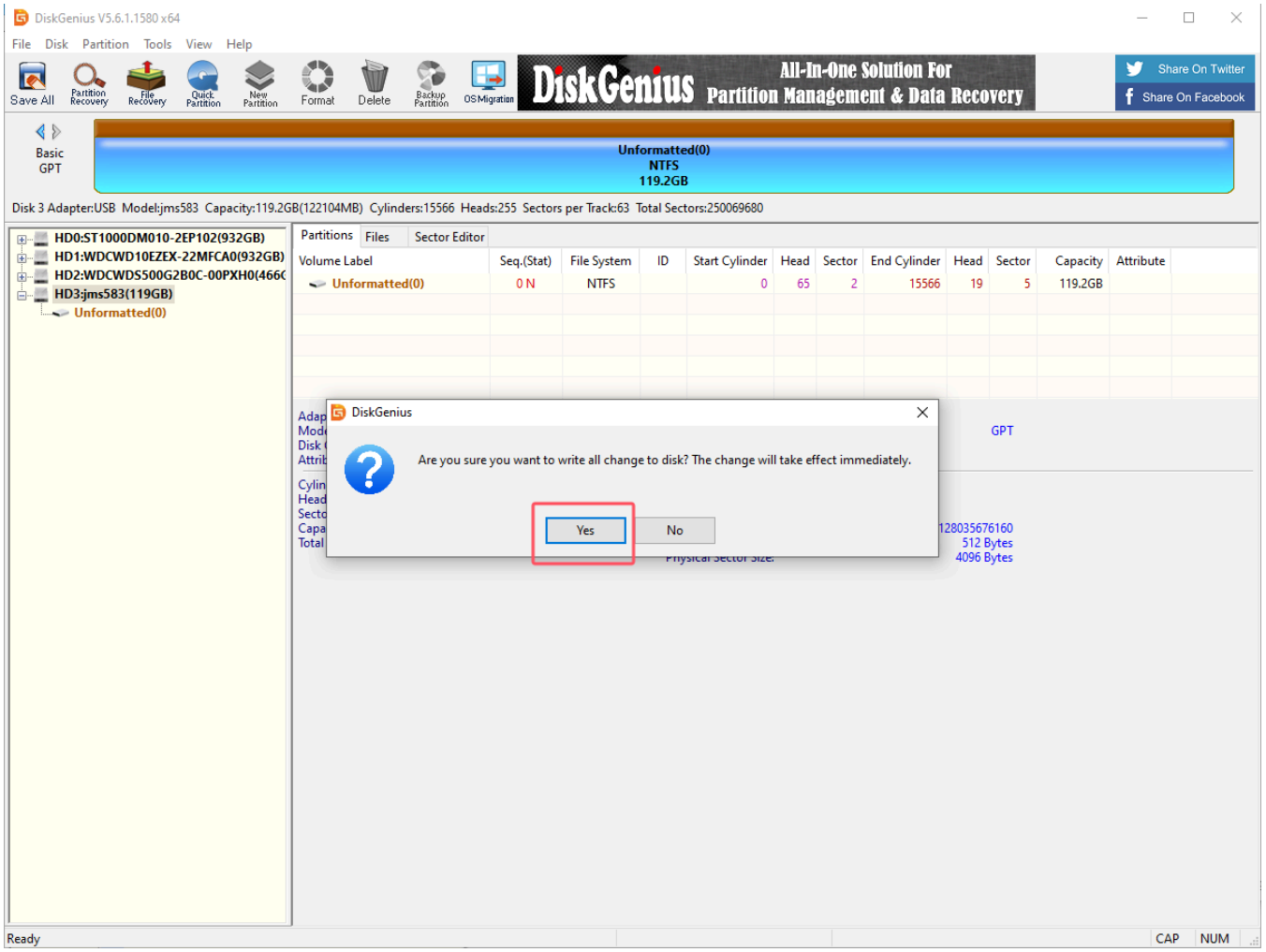
Select the drive letter corresponding to the SSD, and then click New Partition:

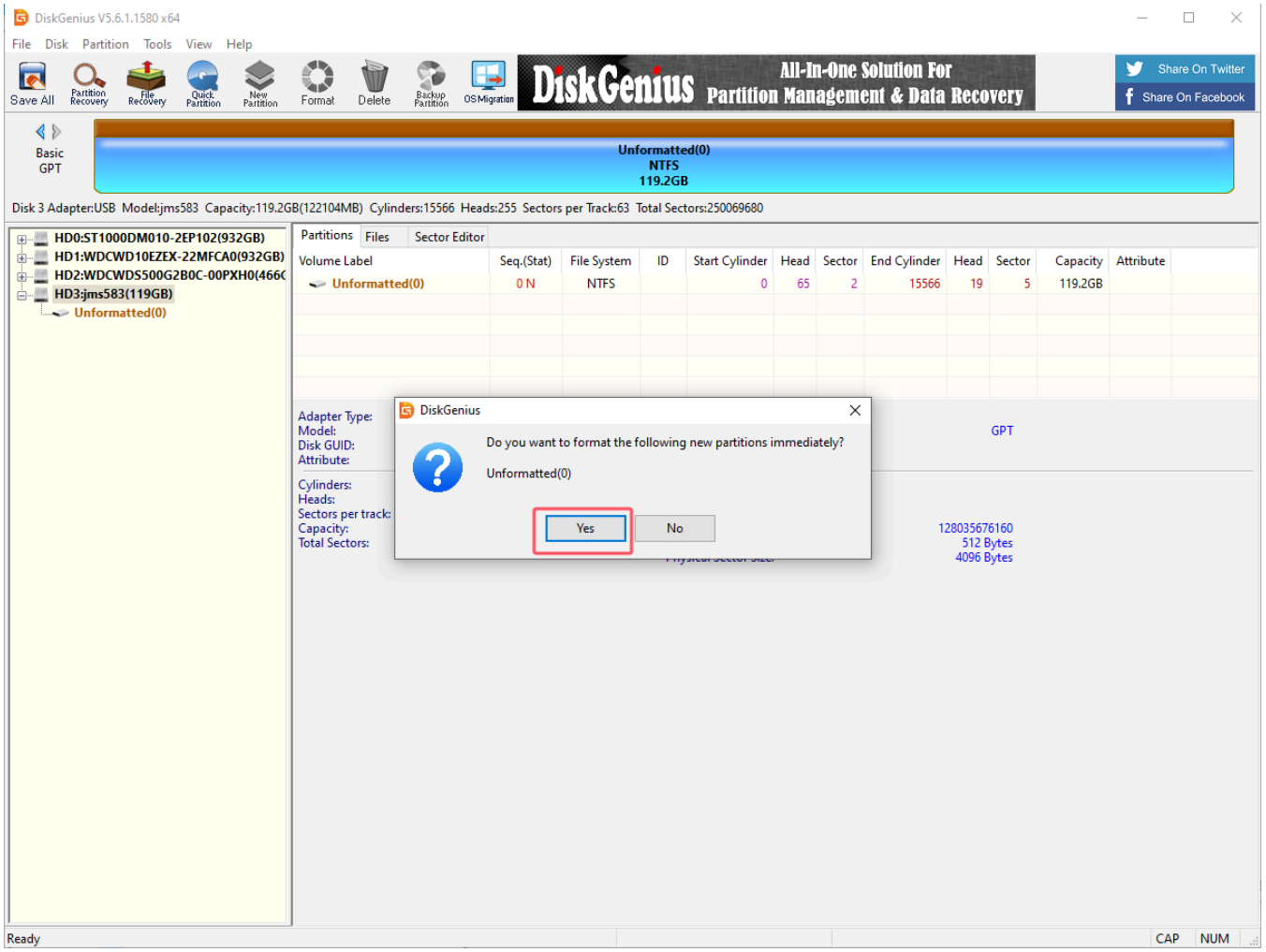

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New Partition
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Delete
Backup Partition
OS Migration
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Share On Twitter
Share On Facebook
Basic GPT
Free
119.2GB
Disk 3 Adapter:USB Model:jms583 Capacity:119.2GB(122104MB) Cylinders:15566 Heads:255 Sectors per Track:63 Total Sectors:250069680
HD0:ST1000DM010-2EP102(932GB)
HD1:WDCWD10EZEX-22MFCA0(932GB)
HD2:WDCWDS500G2B0C-00PXH0(466(
HD3:jms583(119GB)
Partitions
Files
Sector Editor
Volume Label
Seq.(Stat)
File System
ID
Start Cylinder
Head
Sector
End Cylinder
Head
Sector
Capacity
Attribute
Adapter Type: USB
Model: jms583
Disk GUID: 226E3B0F-F9DB-4F9D-A9F9-EB7CE7707966
Attribute: SSD
SN:
Partition Table Style: GPT
Cylinders: 15566
Heads: 255
Sectors per track: 63
Capacity: 119.2GB
Total Sectors: 250069680
Total Bytes: 128035676160
Sector Size: 512 Bytes
Physical Sector Size: 4096 Bytes
Space Type: Free Space
Capacity: 119.2GB
Total Sectors: 250069680
Starting Sector: 0
Total Bytes: 128035676160
CAP
NUM

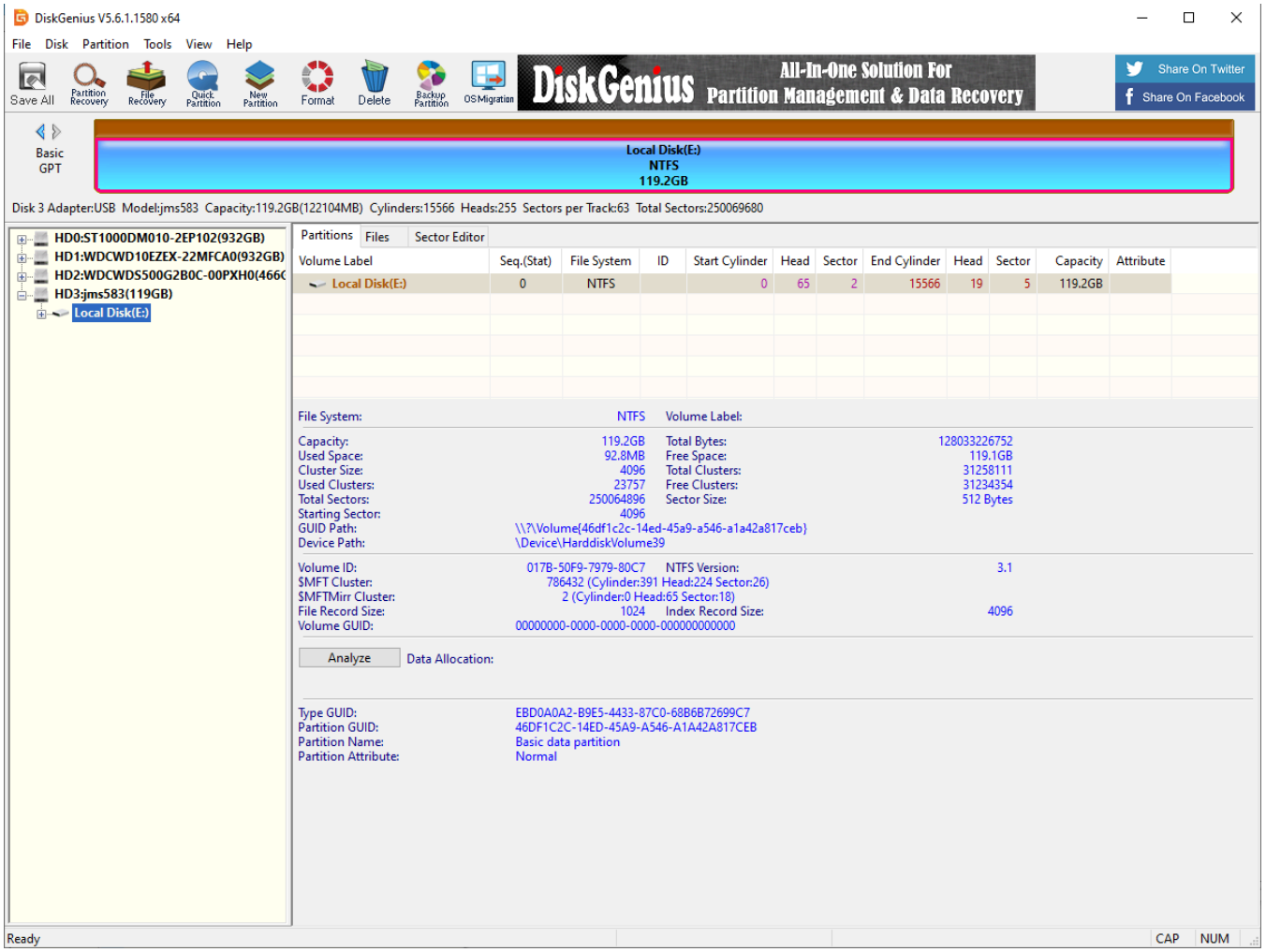

DiskGenius V5.6.1.1580 x64
File Disk Partition Tools View Help
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Basic GPT
Free 119.2GB
Disk 3 Adapter:USB Model:jms583 Capacity:119.2GB(122104MB) Cylinders:15566 Heads:255 Sectors per Track:63 Total Sectors:250069680
HD0:ST1000DM010-2EP102(932GB)
HD1:WDCWD10EZEX-22MFCA0(932GB)
HD2:WDCWDS500G2B0C-00PXH0(466(
HD3:jms583(119GB)
Partitions
Volume Labe
Cylinder Head Sector Capacity Attribute
Adapter Type
Model:
Disk GUID:
Attribute:
GPT
Cylinders:
Heads:
Sectors per tr
Capacity:
Total Sectors:
128035676160
512 Bytes
4096 Bytes
Space Type:
Capacity:
Total Sectors:
Starting Secte
128035676160
Create ESP and MSR partition
Should ESP and/or MSR partition be created before creating new partition?
Create ESP partition
Size of ESP: 300 MB
Create MSR partition
Align to Integral Multiples of Sectors:
2048 Sector (1048576 Byte)
Remark: Microsoft Reserved Partition is used to reserve space ahead of time for other use, and it does not contain any meaningful data itself. Microsoft requires an MSR on every GPT disk, and recommends it to be created as the disk is initially partitioned. It may later automatically shrink on behalf of other partitions, for example during the conversion from basic disk to dynamic disk.
The EFI System Partition(ESP) is a partition on a data storage device that is used by machines that adhere to the Extensible Firmware Interface. It contains the boot loader programs for all operating systems installed (in other partitions) on the device, device driver files (used by the firmware at boot time) for other devices, and system utility programs that are intended to be run before an operating system is booted.
If current disk is installed on the UEFI BIOS computer or may be used under UEFI BIOS computer in the future, an ESP partition should be created.
OK
Cancel
Ready
CAP
NUM


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Basic GPT
Free 119.2GB
Disk 3 Adapter:USB Model:jms583 Capacity:119.2GB(122104MB) Cylinders:15566 Heads:255 Sectors per Track:63 Total Sectors:250069680
HD0:ST1000DM010-2EP102(932GB)
HD1:WDCWD10EZEX-22MFCA0(932GB)
HD2:WDCWDS500G2B0C-00PXH0(466(
HD3:jms583(119GB)
Partitions
Files
Volume Label
End Cylinder
Head
Sector
Capacity
Attribute
Adapter Type:
Model:
Disk GUID:
Attribute:
GPT
Cylinders:
Heads:
Sectors per track:
Capacity:
Total Sectors:
128035676160
512 Bytes
4096 Bytes
Space Type:
Capacity:
Total Sectors:
Starting Sector:
128035676160
Create New Partition
Select Partition Type:
Primary Partition
Extended Partition
Logical Partition
File System Type:
NTFS (MS Basic Data)
File System ID:
07
Size of Partition (0 - 119 GB):
119
GB
Align to Integral Multiples of Sectors:
4096 Sector (2097152 Byte)
Cluster Size:
Default(4096)
Volume Label:
Advanced>>
OK
Cancel
Ready
CAP
NUM


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New Partition
Format
Delete
Backup Partition
OS Migration
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Share On Facebook
Basic GPT
Unformatted(0)
NTFS
119.2GB
Disk 3 Adapter:USB Model:jms583 Capacity:119.2GB(122104MB) Cylinders:15566 Heads:255 Sectors per Track:63 Total Sectors:250069680
HD0:ST1000DM010-2EP102(932GB)
HD1:WDCWD10EZEX-22MFCA0(932GB)
HD2:WDCWDS500G2B0C-00PXH0(466(
HD3:jms583(119GB)
Unformatted(0)
Partitions Files Sector Editor
Volume Label Seq.(Stat) File System ID Start Cylinder Head Sector End Cylinder Head Sector Capacity Attribute
Unformatted(0) 0 N NTFS 0 65 2 15566 19 5 119.2GB
Adapter Type: USB
Model: jms583
Disk GUID: 226E3B0F-F9DB-4F9D-A9F9-EB7CE7707966
Attribute: SSD
SN:
Partition Table Style: GPT
Cylinders: 15566
Heads: 255
Sectors per track: 63
Capacity: 119.2GB
Total Sectors: 250069680
Total Bytes: 128035676160
Sector Size: 512 Bytes
Physical Sector Size: 4096 Bytes
Ready
CAP
NUM


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OS Migration
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All-In-One Solution For
Partition Management & Data Recovery
Share On Twitter
Share On Facebook
Basic
GPT
Unformatted(0)
NTFS
119.2GB
Disk 3 Adapter:USB Model:jms583 Capacity:119.2GB(122104MB) Cylinders:15566 Heads:255 Sectors per Track:63 Total Sectors:250069680
HD0:ST1000DM010-2EP102(932GB)
HD1:WDCWD10EZEX-22MFCA0(932GB)
HD2:WDCWDS500G2B0C-00PXH0(466(
HD3:jms583(119GB)
Unformatted(0)
Partitions
Files
Sector Editor
Volume Label
Seq.(Stat)
File System
ID
Start Cylinder
Head
Sector
End Cylinder
Head
Sector
Capacity
Attribute
Unformatted(0)
0 N
NTFS
0
65
2
15566
19
5
119.2GB
DiskGenius
Are you sure you want to write all change to disk? The change will take effect immediately.
Yes
No
GPT
128035676160
512 Bytes
4096 Bytes
Ready
CAP
NUM


DiskGenius V5.6.1.1580 x64
File Disk Partition Tools View Help
Save All
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File Recovery
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New Partition
Format
Delete
Backup Partition
OS Migration
DiskGenius
All-In-One Solution For
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Share On Twitter
Share On Facebook
Basic
GPT
Unformatted(0)
NTFS
119.2GB
Disk 3 Adapter:USB Model:jms583 Capacity:119.2GB(122104MB) Cylinders:15566 Heads:255 Sectors per Track:63 Total Sectors:250069680
HD0:ST1000DM010-2EP102(932GB)
HD1:WDCWD10EZEX-22MFCA0(932GB)
HD2:WDCWDS500G2B0C-00PXH0(466(
HD3:jms583(119GB)
Unformatted(0)
Partitions
Files
Sector Editor
Volume Label
Seq.(Stat)
File System
ID
Start Cylinder
Head
Sector
End Cylinder
Head
Sector
Capacity
Attribute
Unformatted(0)
0 N
NTFS
0
65
2
15566
19
5
119.2GB
Adapter Type:
Model:
Disk GUID:
Attribute:
GPT
Cylinders:
Heads:
Sectors per track:
Capacity:
Total Sectors:
128035676160
512 Bytes
4096 Bytes
DiskGenius
Do you want to format the following new partitions immediately?
Unformatted(0)
Yes
No
Ready
CAP
NUM

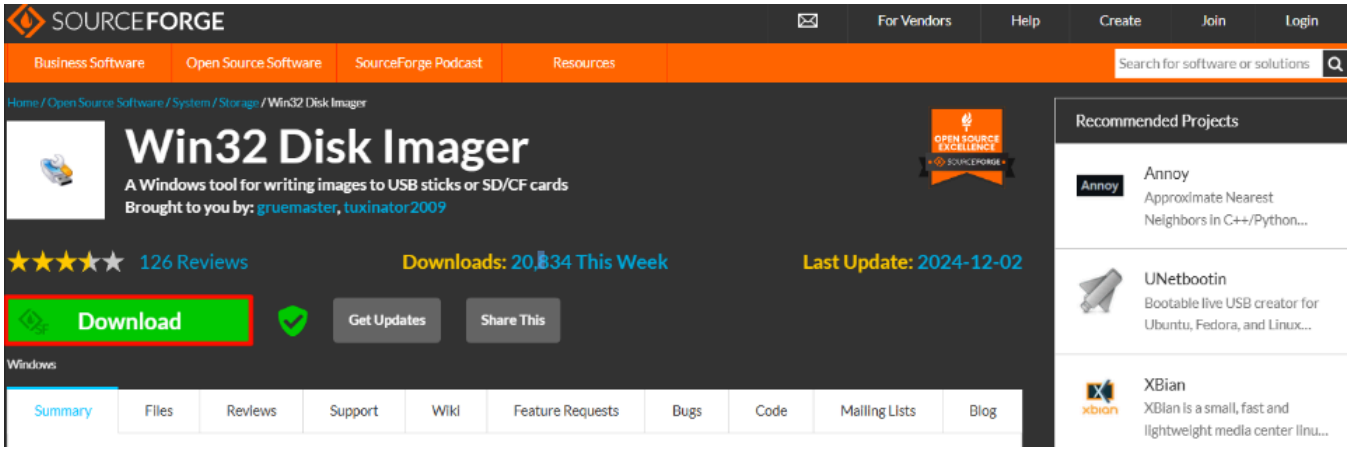


2.2 Restore the factory image

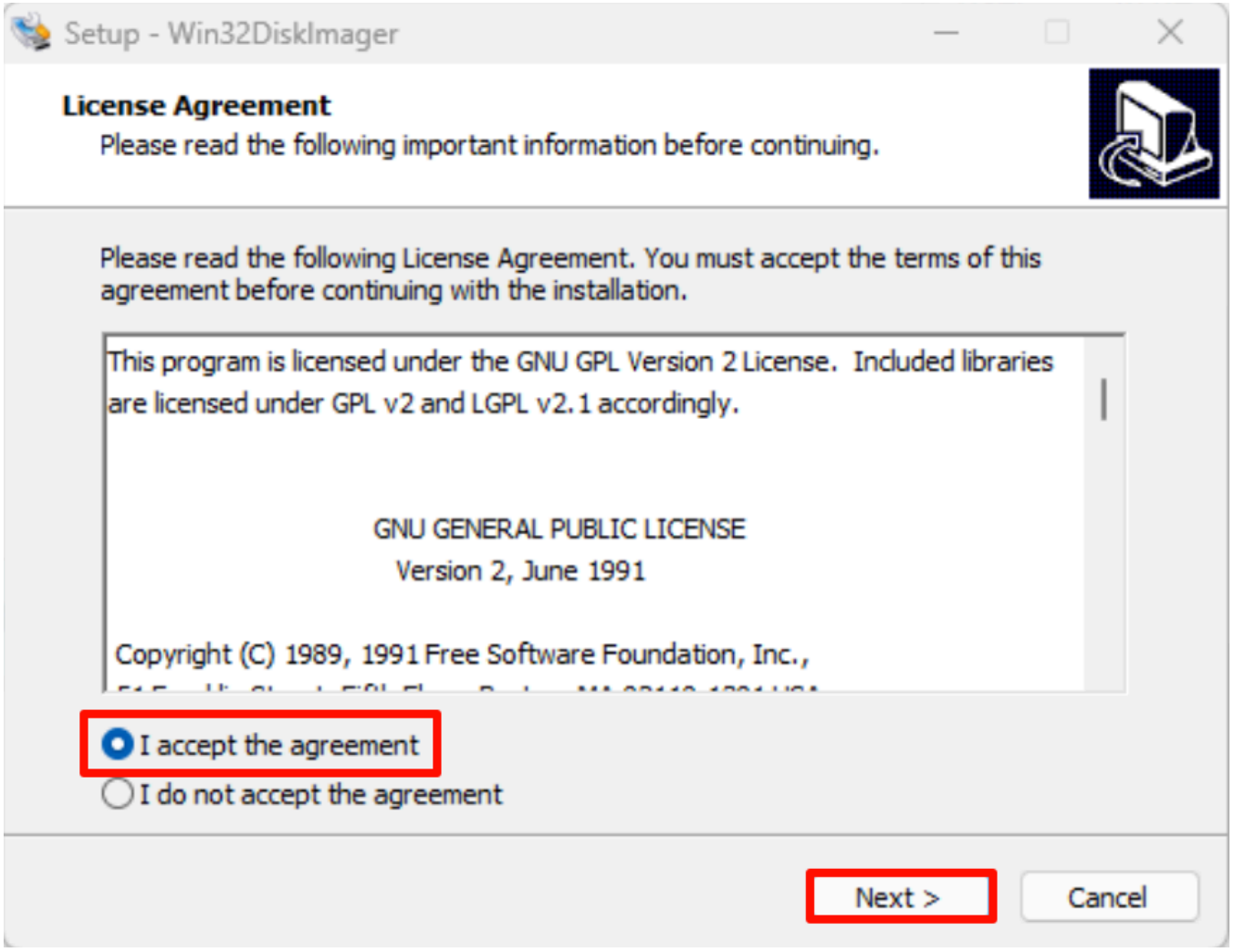
You need to download and decompress the factory image system in the data to the local computer in advance.

2.2.1. Install Win32DiskImager

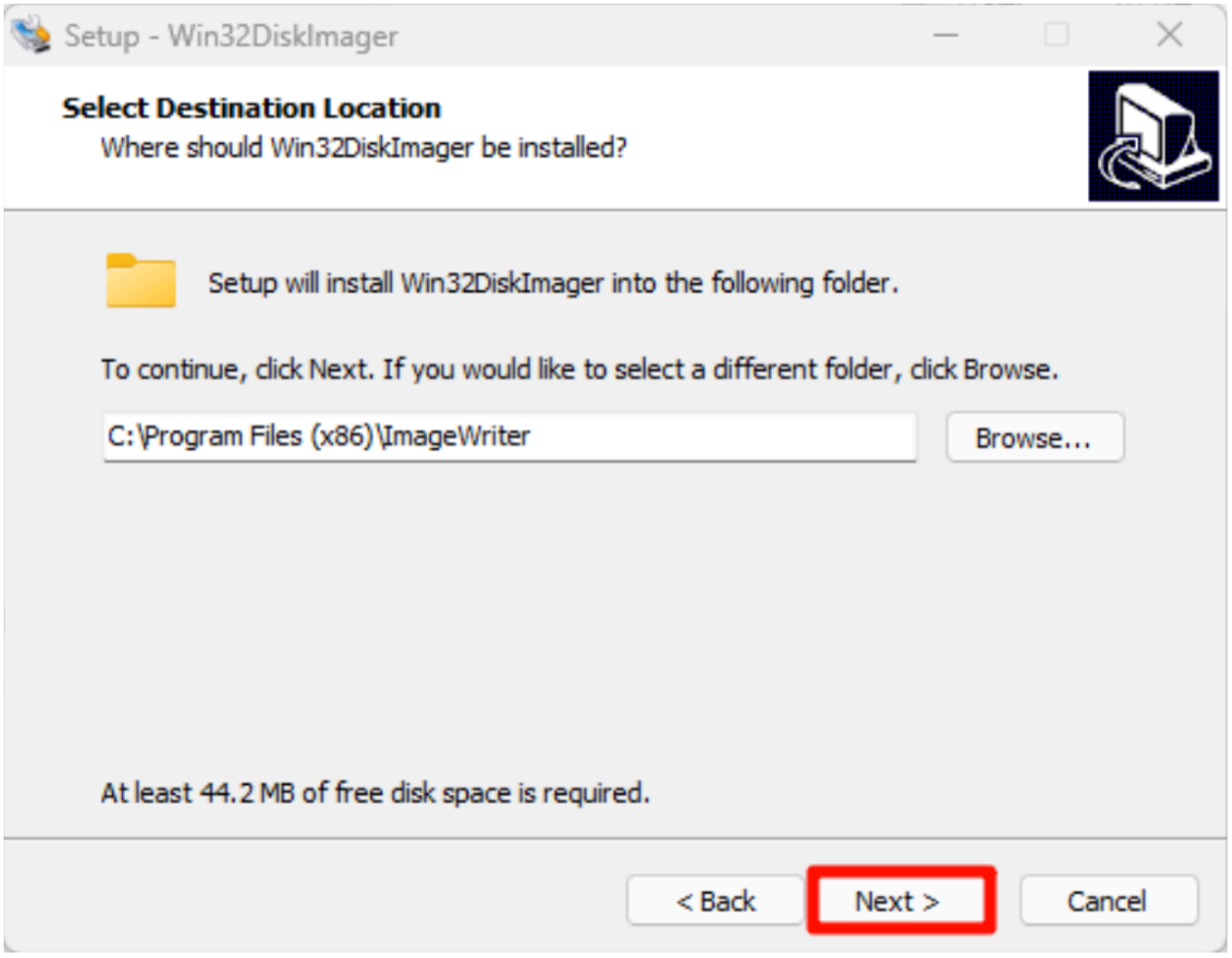
Download URL: https://sourceforge.net/projects/win32diskimager/



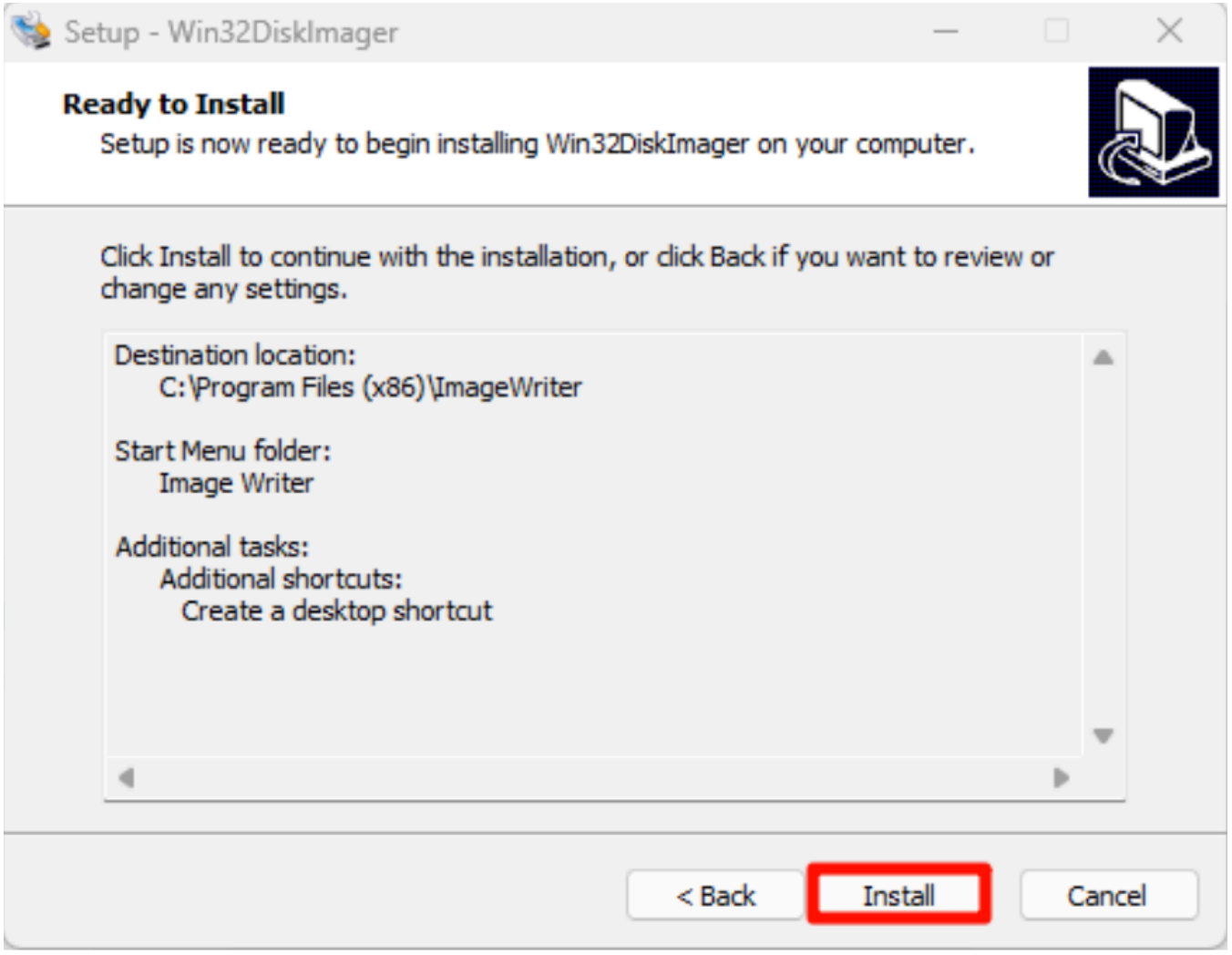
Open the `win32diskimager-1.0.0-install.exe` installation package as an administrator and accept the agreement:

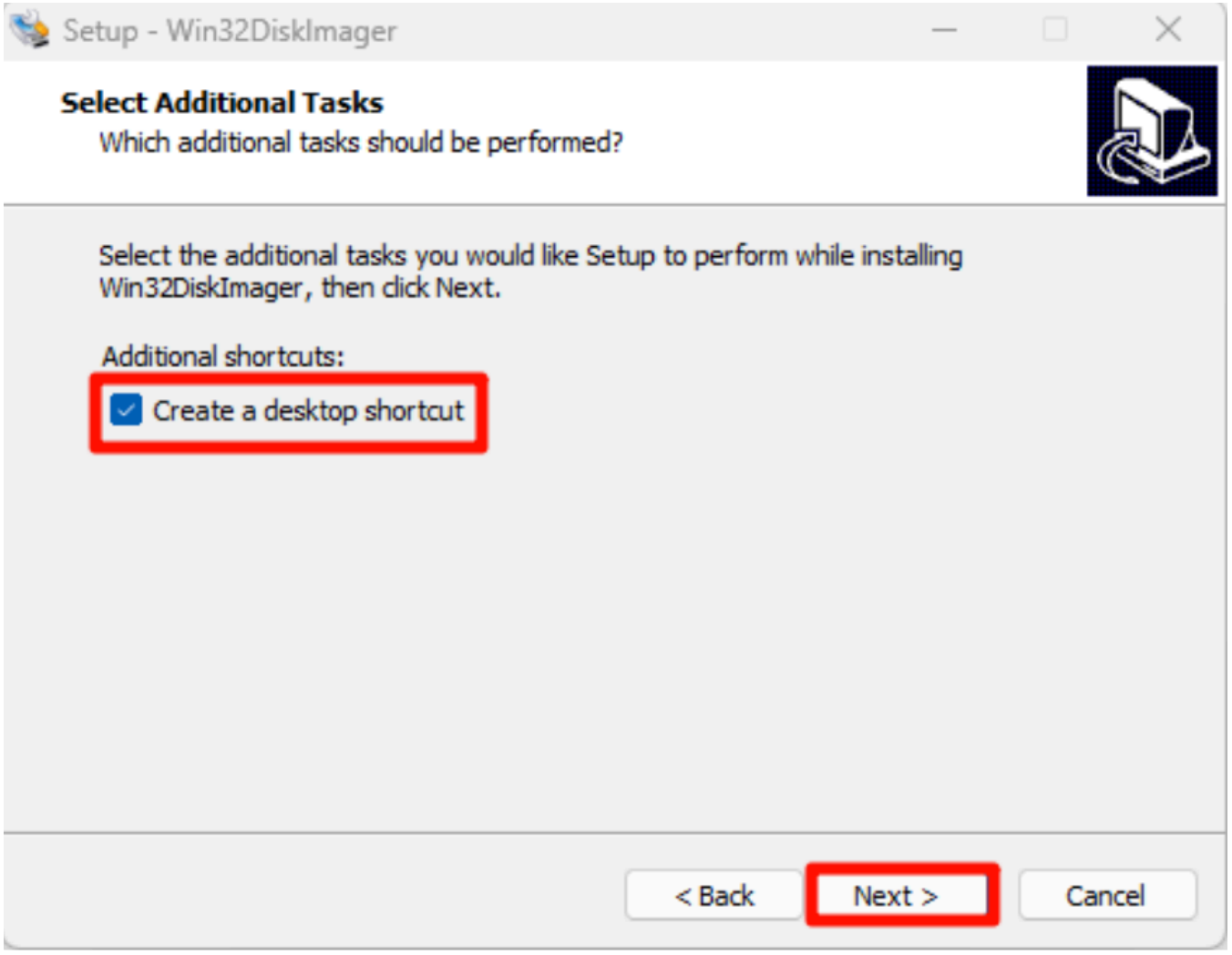


Installation location: The default location is recommended

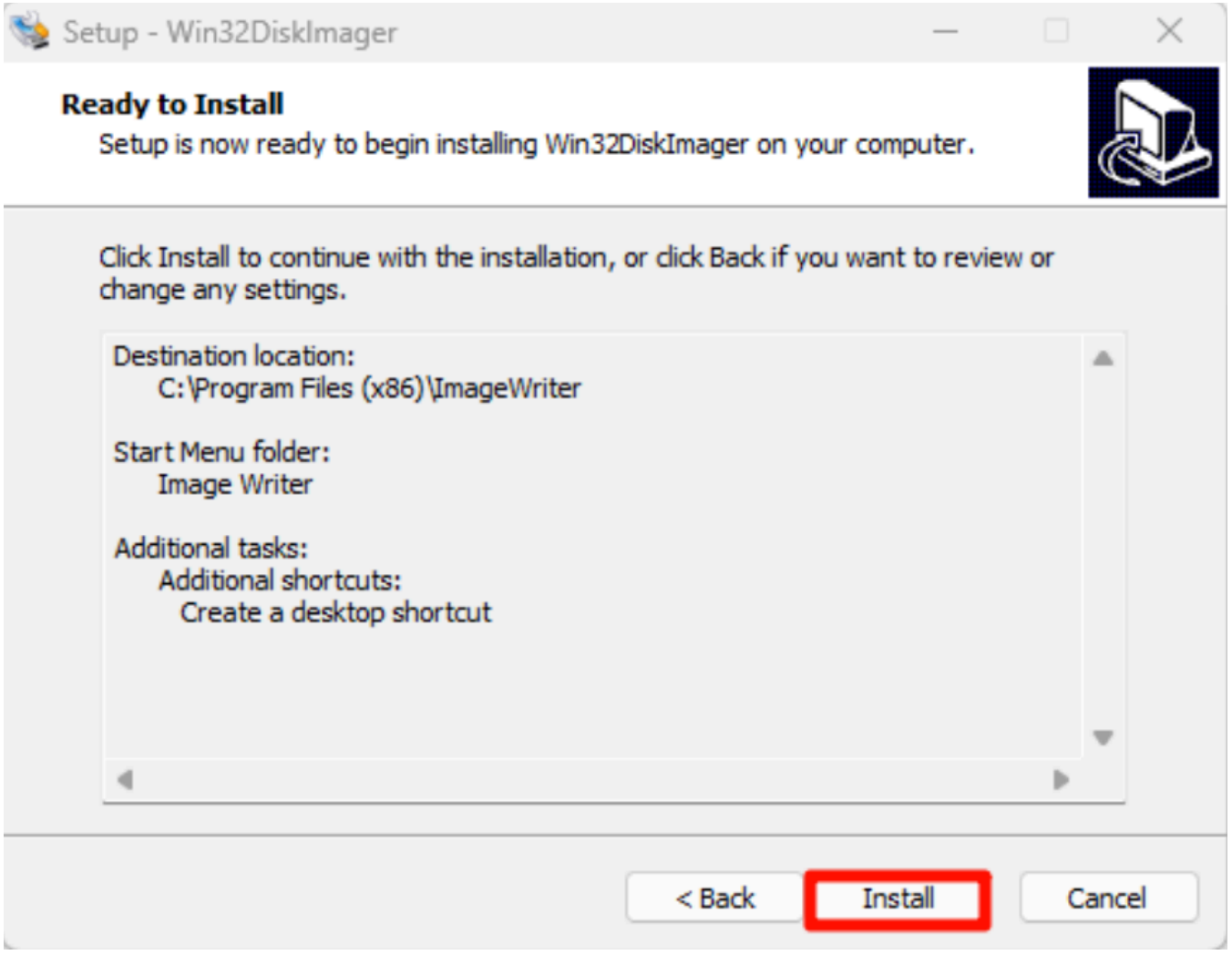


Installation options:

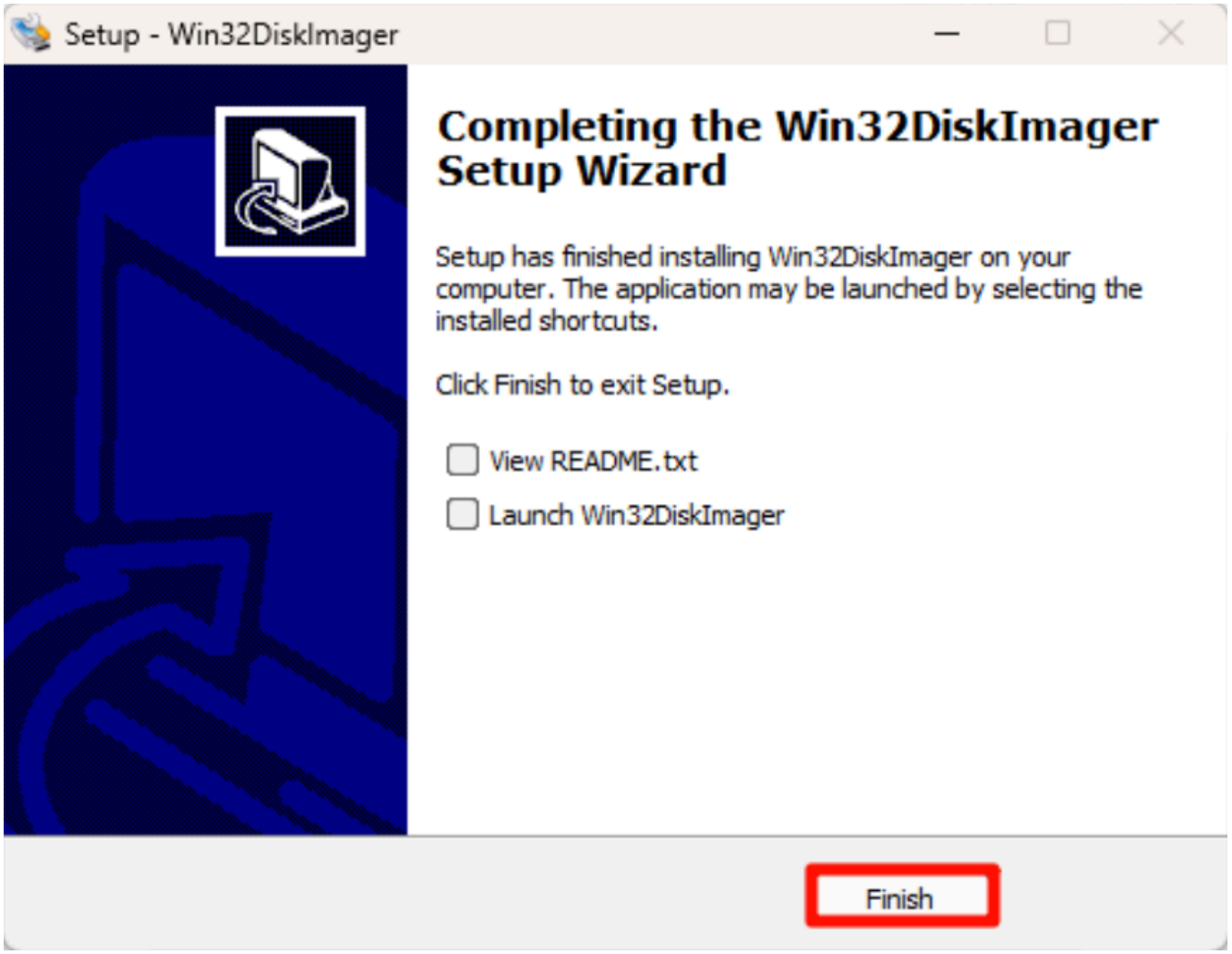

Setup - Win32DiskImager
Ready to Install
Setup is now ready to begin installing Win32DiskImager on your computer.
Click Install to continue with the installation, or click Back if you want to review or change any settings.
Destination location:
C:\Program Files (x86)\ImageWriter
Start Menu folder:
Image Writer
Additional tasks:
Additional shortcuts:
Create a desktop shortcut
< Back
Install
Cancel



Start installation:



Complete installation:

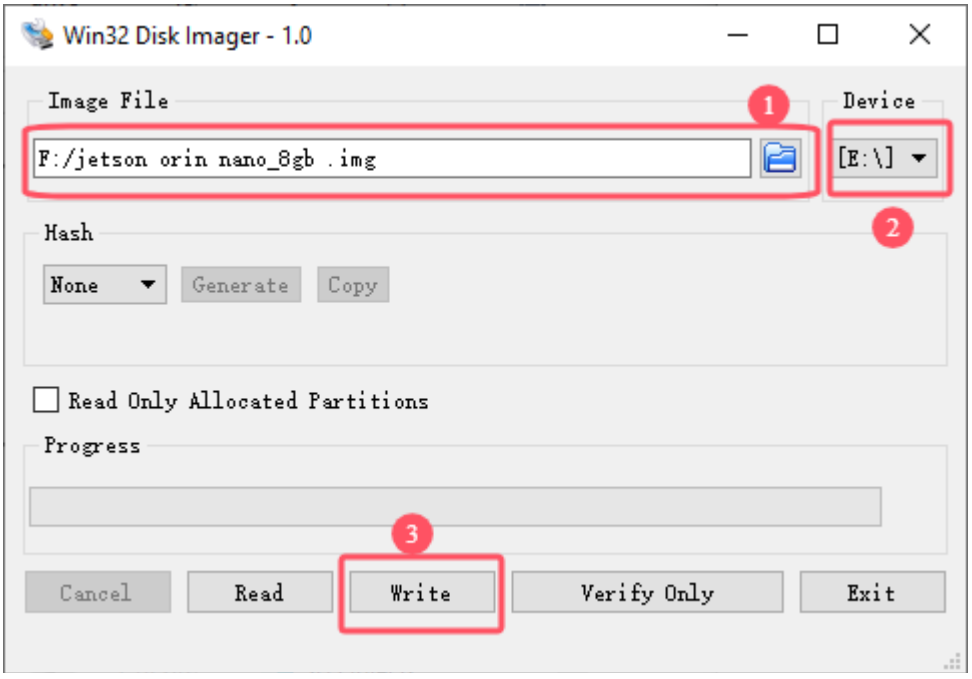


2.2.2 Use Win32DiskImager

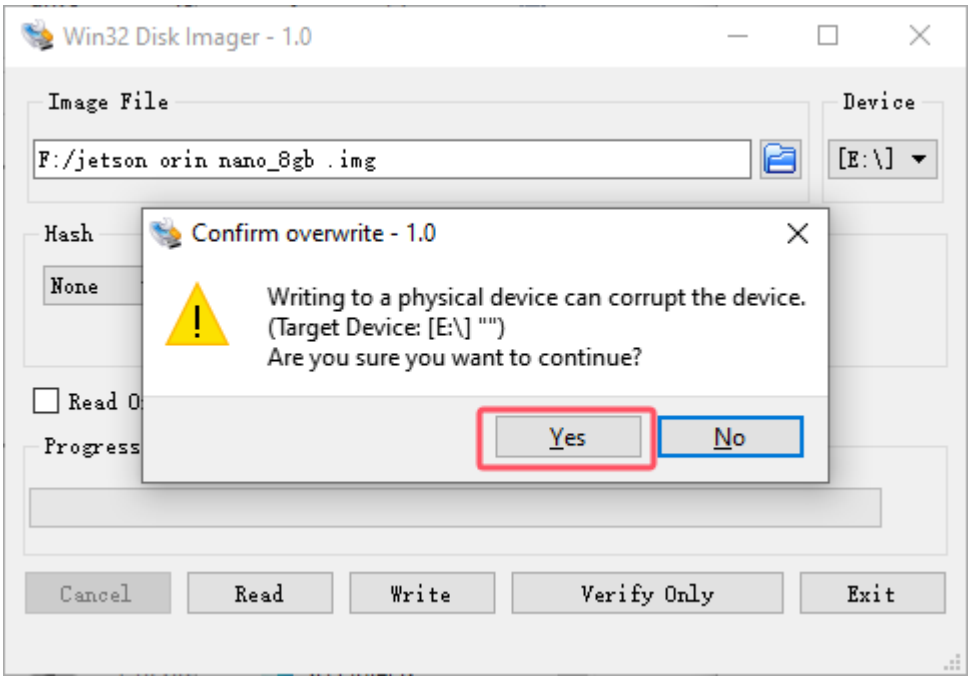
①: Select the factory image file (*.img) in the data

②: Select the drive letter corresponding to the solid-state drive

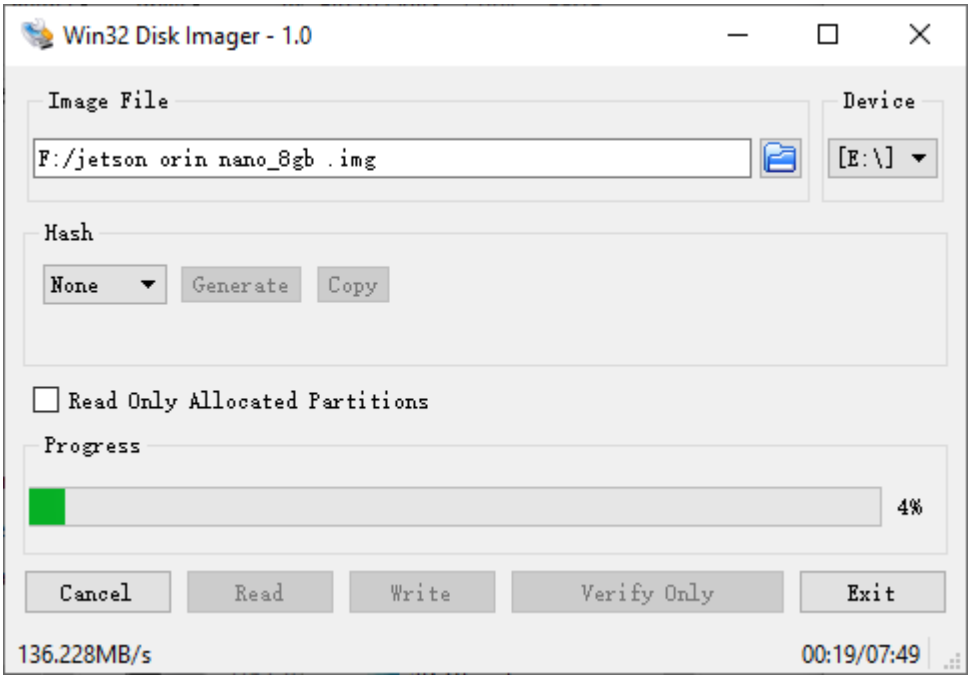
③: Write the factory image to the solid-state drive

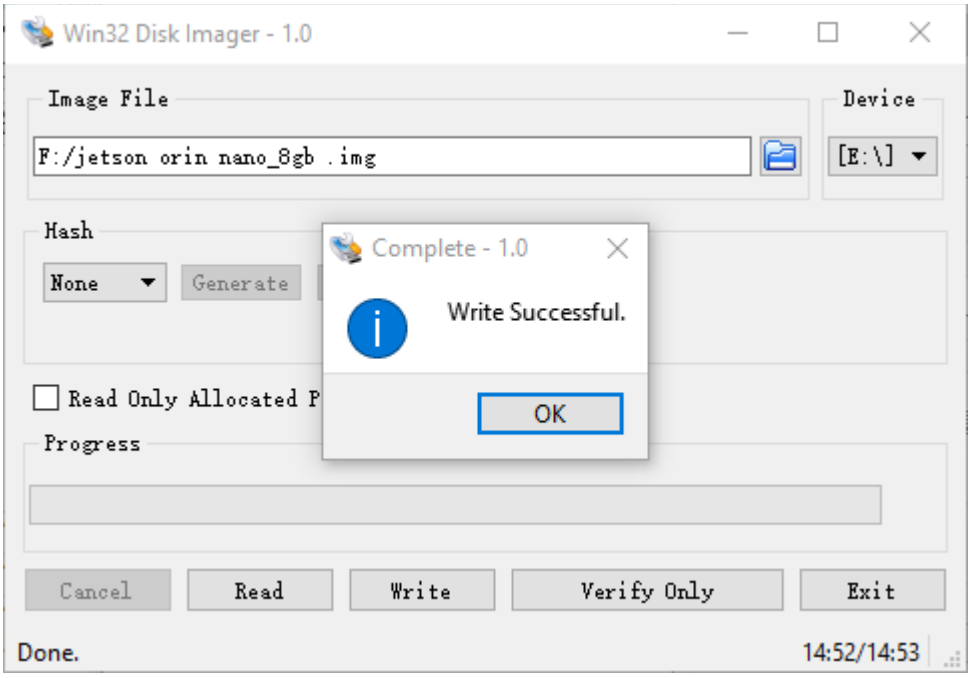


Confirm writing to the system:



Wait for the system to be written successfully:





After the system is written, you can close the program and install the SSD to the Jetson Orin motherboard!